

Thermodynamics

Half-Semester Resource

ENME601 — Weeks 1–6

Compact open-book reference for systems, energy, properties, closed systems, control volumes and the second law

How to use this resource

Start from the problem-type selector, write the complete mass/energy balance, then simplify only with stated assumptions. Use the separate tutorial booklet for source questions, official worked solutions and figures. Use the current *Property Tables 2026* for numerical state properties.

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1 Exam-start decision system

Problem cue	First model choice	Governing start point
Fixed mass, sealed tank, piston-cylinder	Closed system	$Q - W = \Delta U + \Delta KE + \Delta PE$
Inlets/outlets, nozzle, turbine, compressor, valve	Control volume	mass balance first; then SFEE
P, T, x, v, u, h of water/refrigerant	Pure-substance state	classify phase, then select table
Gas far from saturation	Ideal gas	$PV = mRT$ plus appropriate energy relation
Liquid/solid heating	Incompressible	$\Delta u \simeq c\Delta T$; check pressure term in Δh
Engine/refrigerator/heat pump	Second law device	energy relation plus η or COP

Universal workflow

1. Draw the boundary and label mass, heat and work arrows.
2. List knowns/unknowns; convert to absolute pressure and absolute temperature where required.
3. State assumptions (steady, adiabatic, rigid, ideal gas, negligible $\Delta KE/\Delta PE$, etc.).
4. Write unsimplified mass and energy balances before deleting terms.
5. Determine properties from the correct region/table; interpolate only after classification.
6. Substitute with units; check sign, magnitude and physical direction.

2 Week 1 — systems, state and basic properties

2.1 Definitions

A **system** is the matter or region selected for study; the **surroundings** are everything else. The boundary may be fixed or moving.

Type	Mass crosses?	Energy crosses?
Closed/control mass	no	possibly
Open/control volume	yes	possibly
Isolated	no	no

Intensive properties do not scale with mass (T, P, ρ, v); extensive properties do (m, V, U, E). Specific properties divide an extensive property by mass.

$$\rho = \frac{m}{V}, \quad v = \frac{V}{m} = \frac{1}{\rho}.$$

A simple compressible state is fixed by **two independent intensive properties**. In a saturated mixture, P and T are dependent.

2.2 Processes and cycles

Isothermal: $T = \text{const}$; isobaric: $P = \text{const}$; isochoric: $V = \text{const}$; adiabatic: $Q = 0$. Heat and work depend on the path; properties depend only on end states. Over a cycle, every property has zero net change.

2.3 Temperature and pressure

$$T[\text{K}] = T[^\circ\text{C}] + 273.15, \quad P_{\text{abs}} = P_{\text{atm}} + P_{\text{gauge}}, \quad P_{\text{abs}} = P_{\text{atm}} - P_{\text{vac}}.$$

Use absolute P and T in gas and Carnot equations.

Common trap

Steady does not mean uniform. Heat and work are boundary interactions, not properties stored in a system. Celsius temperature differences equal kelvin differences, but absolute-temperature ratios require kelvin.

3 Week 2 — energy transfer, first law and ideal gases

3.1 Stored energy

$$E = U + KE + PE, \quad e = u + \frac{V^2}{2} + gz,$$

$$\Delta KE = \frac{m(V_2^2 - V_1^2)}{2}, \quad \Delta PE = mg(z_2 - z_1).$$

Velocity terms in SI are J/kg; divide by 1000 before adding to kJ/kg.

3.2 Heat, work and sign convention

This resource uses $Q > 0$ into and $W > 0$ by the system:

$$Q - W = \Delta E.$$

For stationary closed systems, $Q - W = \Delta U$. Heat is transfer caused solely by temperature difference. Work includes moving-

boundary, shaft, electrical, spring, lifting and acceleration effects.

$$\dot{W}_{\text{shaft}} = \tau\omega = 2\pi N\tau, \quad W_{\text{spring}} = \frac{1}{2}k(x_2^2 - x_1^2), \quad W_{\text{lift}} = mg\Delta z.$$

3.3 Ideal-gas relations

$$Pv = RT, \quad PV = mRT, \quad R = \frac{R_u}{M}.$$

For fixed mass:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}.$$

With P in kPa and V in m^3 , PV is kJ.

Common trap

$Q = mc\Delta T$ is not the general first law. It is a property-change shortcut under a suitable model, with no phase change. Use c_v for ideal-gas Δu and c_p for Δh .

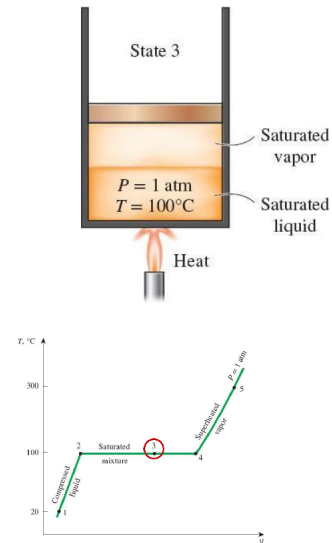
4 Week 3 — pure substances and property tables

4.1 Phase classification



Phase-Change Processes of Pure Substances

- **State 3:** the water is still at 1 atm and 100°C but a portion is saturated liquid, and a portion is saturated vapour and is called a **saturated liquid-vapour mixture** (or just **saturated mixture**).
- A **saturated liquid-vapour mixture** is where the liquid and vapour (gas) phases coexist in equilibrium.
- Temperature = T_{sat} , Pressure = P_{sat}
- The properties of the saturated mixture depend on the amount vaporised (quality x)
- As heat energy is added more saturated liquid will vaporise (boiling continues).
 - Temperature remains constant
 - Specific volume increases significantly



Known at fixed P	Region	Action
$T < T_{sat}(P)$	compressed/subcooled liquid	compressed table or $y \approx y_f(T)$
$T = T_{sat}(P)$	saturated state	another property or quality is required
$T > T_{sat}(P)$	superheated vapour	superheated table

4.2 Saturation and quality

Never interpolate before selecting the correct region and table.

Subscripts: f saturated liquid, g saturated vapour, $fg = g - f$.

$$y_{fg} = y_g - y_f, \quad x = \frac{m_g}{m_f + m_g},$$

$$\boxed{y = y_f + xy_{fg}}, \quad x = \frac{y - y_f}{y_{fg}},$$

for $y \in \{v, u, h, s\}$. Quality is defined only inside the saturation dome and must satisfy $0 \leq x \leq 1$.

4.3 Enthalpy and interpolation

$$h = u + Pv.$$

Linear interpolation between entries (X_1, y_1) and (X_2, y_2) :

$$y = y_1 + \frac{X - X_1}{X_2 - X_1}(y_2 - y_1).$$

4.4 Compressed-liquid approximation

$$y(T, P) \approx y_f(T),$$

$$h(T, P) \approx h_f(T) + v_f(T)[P - P_{sat}(T)].$$

Fast table workflow

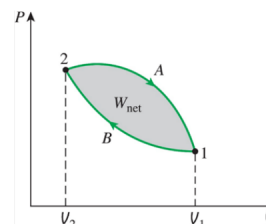
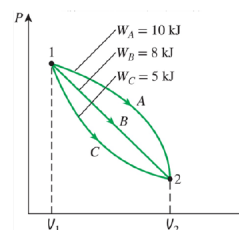
Substance $\rightarrow$ two known intensive properties $\rightarrow$ compare with saturation $\rightarrow$ classify region $\rightarrow$ choose table $\rightarrow$ interpolate $\rightarrow$ sanity-check ordering. A mixture property must lie between its f and g limits.

5 Week 4 — closed-system energy analysis

Moving Boundary Work

- The **ideal moving boundary work** is the **area** under the process path on a $P - V$ diagram.
- For an **expansion process** – work out $\Rightarrow W_b$ is **positive**.
- For a **compression process** – work in $\Rightarrow W_b$ is **negative**.
- For different paths between state 1 and state 2 the **boundary work will be different for each path** (since area under process path is different).
- In a **cycle** the **ideal net work done**, W_{net} is equal to the area enclosed by the process paths (which is equal to the difference between **work done by the system** and **work done on the system**)

work done by the system = area under the expansion curve
work on the system = area under the compression curve



5.1 Boundary work

For quasi-equilibrium piston motion:

$$W_b = \int_{V_1}^{V_2} P dV$$

Expansion gives positive work by the system; compression gives negative work. Work is the area under the process path.

Model	Boundary work
Constant P	$P(V_2 - V_1)$
Rigid tank	0
Isothermal ideal gas	$mRT \ln(V_2/V_1)$
Polytropic, $n \neq 1$	$\frac{P_2 V_2 - P_1 V_1}{1 - n}$
Rev. adiabatic ideal gas	polytropic with $n = k$

For a polytropic process $PV^n = \text{const.}$ For a reversible adiabatic ideal-gas process $PV^k = \text{const.}$ Adiabatic alone is insufficient for the latter relation.

5.2 Closed-system balance

$$Q - W = \Delta U + \Delta KE + \Delta PE.$$

Stationary system: $Q - W = m(u_2 - u_1)$. At constant pressure with boundary work as the only work:

$$Q = \Delta H = m(h_2 - h_1).$$

5.3 Specific heats

$$c_v = \left(\frac{\partial u}{\partial T} \right)_v, \quad c_p = \left(\frac{\partial h}{\partial T} \right)_p$$

Ideal gas:

$$\Delta u = \int c_v(T) dT \approx c_{v,avg} \Delta T,$$

$$\Delta h = \int c_p(T) dT \approx c_{p,avg} \Delta T, \quad c_p - c_v = R, \quad k = \frac{c_p}{c_v}.$$

Incompressible liquid/solid:

$$\Delta u \approx c \Delta T, \quad \Delta h \approx c \Delta T + v \Delta P.$$

Common trap

Constant pressure is not isothermal. Isothermal is not constant pressure. A rigid tank has $W_b = 0$ but may still have heat, shaft or electrical work.

6 Week 5 — control volumes and steady-flow devices

Energy Analysis of Steady Flow Systems

- For a **steady flow process** the total energy of the control volume, remains constant i.e. **the rate of change of energy of the control volume (system) = 0**.
Writing the energy balance equation in rate form:

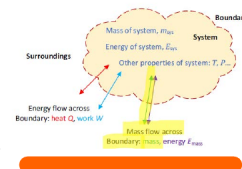
$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\text{rate of net energy transfer by heat, work and mass}} = \underbrace{\cancel{\frac{dE_{system}}{dt}}}_{\text{rate of change in internal kinetic, potential, etc energies}} \quad \text{open system}$$

- Thus, the rate of energy entering the control volume in all forms (**heat, work and mass transfer**) must be equal to the rate of energy leaving it

$$\underbrace{\dot{E}_{in}}_{\text{rate of net energy transfer by heat, work and mass}} = \underbrace{\dot{E}_{out}}_{\text{rate of net energy transfer by heat, work and mass}}$$

This can be written as:

$$\begin{aligned} \dot{Q}_{in} + \dot{W}_{in} + \sum \dot{E}_{mass,in} &= \dot{Q}_{out} + \dot{W}_{out} + \sum \dot{E}_{mass,out} \\ \Rightarrow \dot{Q}_{in} + \dot{W}_{in} + \sum \dot{m}_{in} e_{flow,in} &= \dot{Q}_{out} + \dot{W}_{out} + \sum \dot{m}_{out} e_{flow,out} \end{aligned}$$



6.1 Mass and stream energy

$$\dot{V} = V_{avg} A, \quad \boxed{\dot{m} = \rho \dot{V} = \rho V_{avg} A = \frac{\dot{V}}{v}}$$

General mass balance:

$$\sum \dot{m}_{in} - \sum \dot{m}_{out} = \frac{dm_{CV}}{dt}$$

At steady state, $\sum \dot{m}_{in} = \sum \dot{m}_{out}$. Flow work Pv combines with u to form enthalpy, so a stream carries

$$e_{flow} = h + \frac{V^2}{2} + gz$$

6.2 Steady-flow energy equation

Using heat in and work out as positive:

$$\dot{Q} - \dot{W} = \sum_{out} \dot{m} \left(h + \frac{V^2}{2} + gz \right) - \sum_{in} \dot{m} \left(h + \frac{V^2}{2} + gz \right)$$

One inlet and outlet:

$$\boxed{\dot{Q} - \dot{W} = \dot{m} \left[\Delta h + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1) \right]}$$

Write mass balance first. Simplify only after inspecting the device.

6.3 Device selector

Device	Typical assumptions	Dominant result
Nozzle/diffuser	steady, adiabatic, no shaft work, $\Delta PE \simeq 0$	$h_1 + V_1^2/2 = h_2 + V_2^2/2$
Turbine	adiabatic, $\Delta KE, \Delta PE \simeq 0$	$\dot{W}_{out} = \dot{m}(h_1 - h_2)$
Compressor	adiabatic, $\Delta KE, \Delta PE \simeq 0$	$\dot{W}_{in} = \dot{m}(h_2 - h_1)$
Pump, incompressible	small ΔT	$w_{in} \simeq v(P_2 - P_1)$
Throttle	adiabatic, no work, negligible $\Delta KE, \Delta PE$	$h_1 = h_2$
Mixing chamber	adiabatic, no work	mass balance plus $\sum \dot{m} h_{in} = \sum \dot{m} h_{out}$
Heat exchanger, whole	externally adiabatic, no work	hot-stream loss = cold-stream gain
Filling/emptying tank	transient, not steady	integrated mass and uniform-flow energy balances

Common trap

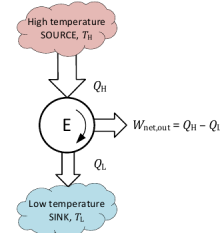
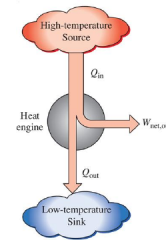
A constant inlet flow does not make a filling tank steady: mass accumulates. Throttling is isenthalpic, not necessarily isothermal or isentropic. Distinguish speed V from volume flow rate $\dot{V}$.

7 Week 6 — second law, engines, refrigeration and Carnot limits



Heat Engines

- Work can be converted directly and completely to heat (friction) but **converting heat to work requires a special device called a heat engine**.
- Heat engines are characterised by
 - They receive heat, Q_H from a high-temperature source (at T_H). (combustion of fuel, solar energy, nuclear reactor etc).
 - They convert part of this heat to work, $W_{\text{net,out}}$ (e.g. rotating shaft).
 - They reject remaining waste heat, Q_L to a low-temperature sink (at T_L). (Usually the atmosphere, lake etc).
 - They operate on a cycle.
- Heat engines and other cyclic devices usually involve a fluid called the **working fluid**. Heat is transferred to and from the working fluid as it undergoes the cycle.
- Note: $Q_H = Q_{\text{in}}$, $Q_L = Q_{\text{out}}$ (and are all magnitudes).



7.1 Heat engine

$$W_{\text{net,out}} = Q_H - Q_L, \quad \eta_{\text{th}} = \frac{W_{\text{net,out}}}{Q_H} = 1 - \frac{Q_L}{Q_H}.$$

A cyclic engine must reject heat; a 100% heat engine violates Kelvin–Planck.

$$\boxed{COP_{R,\text{rev}} = \frac{T_L}{T_H - T_L}}, \quad \boxed{COP_{HP,\text{rev}} = \frac{T_H}{T_H - T_L}}.$$

A real device cannot exceed the reversible limit for the same reservoirs.

7.2 Refrigerator and heat pump

$$W_{\text{net,in}} = Q_H - Q_L, \quad \boxed{COP_R = \frac{Q_L}{W_{\text{net,in}}}}, \quad \boxed{COP_{HP} = \frac{Q_H}{W_{\text{net,in}}} = COP_R + 1}.$$

A COP above one is normal because work moves additional heat; COP is not heat-engine efficiency.

7.3 Carnot limits

For reservoirs at absolute temperatures T_H and T_L :

$$\boxed{\eta_{\text{th,rev}} = 1 - \frac{T_L}{T_H}},$$

7.4 Reversibility

Irreversibilities include friction, unrestrained expansion, mixing, heat transfer across a finite temperature difference, electrical resistance, combustion, turbulence and shock. Reversible devices provide maximum expansion work, minimum compression work and performance limits.

Common trap

Carnot equations require kelvin. Do not confuse Q_H (hot-side heat) with Q_L (cold-side heat). Identify the desired output before choosing efficiency, refrigerator COP or heat-pump COP.

8 Compact formula and unit checks

Property and gas checks

- $1 \text{ kPa m}^3 = 1 \text{ kJ}$.
- $R_u = 8.31447 \text{ kJ}/(\text{kmol K})$ and $R = R_u/M$.
- Quality only in two-phase region.
- Use one consistent property-table reference set.

Balance checks

- Closed system: no mass crosses.
- Control volume: mass balance before energy balance.
- Stationary is not automatically adiabatic.
- Cross out terms only with a physical reason.

Sign checks

- This resource: Q into and W by are positive.
- Expansion boundary work positive; compression negative.
- Turbine normally work out; compressor normally work in.
- Heat engine: $Q_H = W_{\text{net,out}} + Q_L$.

Current local companions

- *ENME601 Formula Sheet 2026.pdf*
- *Property Tables 2026.pdf*
- Separate *Thermodynamics Tutorial Booklet Weeks 1–6*
- Mid-semester practice tests and answers in the assessment index

9 Source provenance

Built from the local 2026 Lectures 1–6, current tutorial sets, formula sheet and property tables. Recheck numerical state properties against *Property Tables 2026*.